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Operational Plan: A Heterogeneous Federation as Proof of the Mission-Ready Architecture

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Abstract

The Mission-Ready AI Decalogy's first seven papers establish the architecture of governed agentic operation: the HGC³AE² framework, the epistemic constraints under which agentic systems operate, the seven-layer edge doctrine, the cognitive node, the Skipjack Protocol, the operating model, and the agentic substrate. The eighth paper specifies the comprehension instrument by which a mission-ready system maintains the self-knowledge required to remain inside its governance envelope. What is absent, after these eight papers, is the empirical demonstration that the architecture survives contact with production operation. This paper closes that gap.

This paper documents an actually-deployed heterogeneous federation — an on-premises GPU cluster operated under K3s, a hyperscaler-managed inference and storage surface, and a sovereign-jurisdiction bare-metal capacity tier — governed by a single Skipjack-conformant operating layer. It describes what the federation runs, where workload is routed and why, which failure modes have been encountered, how Skipjack's four enforcement mechanisms held under contact, and what the unit economics look like compared to single-hyperscaler deployment of the same capability set.

The principal observation is not that any single substrate is best for agentic operation — none is — but that a governed federation across heterogeneous substrates can satisfy the architecture's requirements at unit economics that single-hyperscaler deployment cannot match, for the workload classes that mission-ready agentic operation typically demands: steady-state inference, persistent memory, governance overhead, and forward-deployed DDIL-tolerant nodes.

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The Mission-Ready AI Decalogy's first seven papers establish the architecture of governed agentic operation: the HGC³AE² framework, the epistemic constraints under which agentic systems operate, the seven-layer edge doctrine, the cognitive node, the Skipjack Protocol, the operating model, and the agentic substrate.[1][2][3][4][5][6][7] The eighth paper specifies the comprehension instrument by which a mission-ready system maintains the self-knowledge required to remain inside its governance envelope.[8] What is absent, after these eight papers, is the empirical demonstration that the architecture survives contact with production operation. This paper closes that gap.

This paper documents an actually-deployed heterogeneous federation — an on-premises GPU cluster operated under K3s, a hyperscaler-managed inference and storage surface, and a sovereign-jurisdiction bare-metal capacity tier — governed by a single Skipjack-conformant operating layer. It describes what the federation runs, where workload is routed and why, which failure modes have been encountered, how Skipjack's four enforcement mechanisms held under contact, and what the unit economics look like compared to single-hyperscaler deployment of the same capability set.

The argument is empirical rather than theoretical. The Mission-Ready architecture is not a future-tense proposal; it is operationally viable today, on commercially available substrates, with public cloud and on-premises infrastructure that organizations already procure. The principal observation is not that any single substrate is best for agentic operation — none is — but that **a governed federation across heterogeneous substrates can satisfy the architecture's requirements at unit economics that single-hyperscaler deployment cannot match**, for the workload classes that mission-ready agentic operation typically demands: steady-state inference, persistent memory, governance overhead, and forward-deployed DDIL-tolerant nodes.

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1 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869285.

2 Justin H. Kuiper, CISSP, *Epistemic Constraints* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869287.

3 Justin H. Kuiper, CISSP, *Edge AI Doctrine* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869289.

4 Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869307.

5 Justin H. Kuiper, CISSP, *The Skipjack Protocol* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869293.

6 Justin H. Kuiper, CISSP, *The Operating Model for Agentic Teams* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869313.

7 Justin H. Kuiper, CISSP, *The Agentic Substrate* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869291.

8 Forthcoming MR-P8 *The Comprehension Instrument for Agentic Systems* (v1.0-preprint, in concurrent drafting; yks-build#98).

The paper draws on operational telemetry, governance event logs, persistence-tier survival records, and a cost ledger compiled from invoicing and capacity-utilization data over the federation's operating window. It is not an argument from analogy or projection; it is a case study in the technical sense.

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1. INTRODUCTION

The presumption that an agentic operating model belongs on a single hyperscaler is a holdover from an earlier era of cloud computing economics. It made sense when commodity compute was the binding constraint, when accelerator allocation was a non-issue, when sovereignty regimes were peripheral concerns, and when the workload mix on cloud was dominated by transactional services with bursty demand profiles. It makes substantially less sense for an agentic operating model in 2026, where steady-state inference dominates the workload mix, accelerator allocation is the binding upstream constraint on capacity at every hyperscaler, sovereignty regimes determine where workloads may legally execute, and the persistent state agentic systems maintain creates portability friction that the single-hyperscaler default does not relieve.[9][10]

The Managing Agentics Ops series (in concurrent draft) treats each layer of this shift at depth: capacity sourcing as distinct from cloud identity,[11] heterogeneous compute routing as a multi-policy problem,[12] and the capex/opex inflection that determines when bare-metal capacity beats cloud routing.[13] What that series does not do — what the present paper does — is demonstrate the architecture *in operation*, on real workloads, with measured outcomes. The argument here is that the architecture works, not that it might work.

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9 Michael Armbrust et al., "A View of Cloud Computing," *Communications of the ACM* 53, no. 4 (2010): 50–58, https://doi.org/10.1145/1721654.1721672.

10 Deepak Narayanan et al., "Efficient Large-Scale Language Model Training on GPU Clusters Using Megatron-LM," *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC '21)*, 2021, https://doi.org/10.1145/3458817.3476209.

11 Forthcoming MAO-P1 *Capacity Sourcing* (v0.1-seed, in concurrent drafting; yks-build#92).

12 Forthcoming MAO-P2 *Heterogeneous Compute Routing* (v0.1-seed, in concurrent drafting; yks-build#93).

13 Forthcoming MAO-P3 *The Capex/Opex Inflection* (v0.1-seed, in concurrent drafting; yks-build#94).

The federation described in this paper is the **Non Sequitur Robo Stack**, comprised of three substrate tiers:

An **on-premises GPU cluster (Omen)**, operated under K3s,[14] hosting steady-state inference, persistent memory, governance overhead, and the agentic substrate runtime.

A **hyperscaler-managed surface** (AWS), providing managed accelerator capacity for burst inference, object storage for shared artifacts, and ingress for hyperscaler-fronted external services.

A **sovereign bare-metal tier** (OCI), providing capacity in a sovereignty regime distinct from the primary on-premises and managed-cloud tiers, plus egress paths that satisfy data-residency constraints the other tiers cannot.

The three tiers are governed by a single Skipjack-conformant operating layer that routes workload between them under HGC^3AE^2 governance and reports back into a unified persistence and observability plane.[15][16] The operating layer does not paper over the heterogeneity; it treats heterogeneity as a first-class property and routes accordingly.

The remainder of this paper proceeds in the canonical Mission-Ready shape. Section 2 identifies the five operational failure modes that single-hyperscaler deployment of mission-ready agentic operation has been observed to produce. Section 3 specifies the federation as built. Section 4 treats federation operation as a continuous process, with workload routing under HGC^3AE^2 as the central rhythm. Section 5 walks Skipjack enforcement under federation conditions. Section 6 returns the analysis to the HGC^3AE^2 framework and draws implications for organizations preparing to adopt a similar architecture.

The Mission-Ready architecture has carried, through the first eight papers, a substantial theoretical load. This paper is the moment at which that load is transferred to operational ground.

2. PROBLEM SET: OPERATIONAL FAILURES OF THE SINGLE-HYPERSCALER DEFAULT

Single-hyperscaler deployment of mission-ready agentic operation produces five operational failure modes that recur across observed deployments. They are structural in the same sense that the failure modes identified in earlier papers were structural: they do not reflect inadequate engineering on any single hyperscaler's part. They reflect the mismatch between agentic operating workloads and the architectural assumptions that hyperscaler offerings were built around.

2.1 CAPACITY IDENTITY CONFUSION

The first operational failure is **capacity identity confusion**: the operating model treats *which hyperscaler the workload is invoiced against* as equivalent to *where the capacity that fulfills the workload actually comes from*. The Managing Agentics Ops series develops this

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14 Brendan Burns et al., "Borg, Omega, and Kubernetes," *Communications of the ACM* 59, no. 5 (2016): 50–57, https://doi.org/10.1145/2890784.

15 Justin H. Kuiper, CISSP, *The Skipjack Protocol* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869293.

16 Justin H. Kuiper, CISSP, *The Agentic Substrate* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869291.

confusion in depth and locates its origins in the six-layer capacity stack that determines what a cloud account can actually allocate.[17] In operational terms, the failure surfaces when accelerator allocation tightens, sovereignty regimes shift, or regional power constraints bind: the multi-region failover instrument that the single-hyperscaler default relies on does not address the constraint, because the constraint operates upstream of the account abstraction.

The case-study evidence for this failure is unambiguous. During the accelerator generation transition that bound across Q2–Q3 2025, three reference workloads in the federation were observed to fail over from one hyperscaler region to another to a third, with each transition exposing the same underlying allocation constraint at the next region in line.[18] The on-premises tier was unaffected, not because it was technically superior, but because its capacity was sourced under a different supply-chain relationship altogether.

2.2 PERSISTENCE-TIER LOCK-IN

The second operational failure is **persistence-tier lock-in**. Agentic systems maintain substantial persistent state: conversation memory, governance event logs, intermediate artifacts, retrieval indices.[19] When this persistent state lives in a single hyperscaler's persistence services — object stores, managed databases, vector indices — the cost of moving the agentic system off that hyperscaler is dominated not by compute portability (which is comparatively cheap) but by data portability. Egress fees, schema translation, index reconstruction, and consistency verification each impose costs that scale with the volume of accumulated state.[20]

The federation architecture addresses this by holding the authoritative persistence tier on the on-premises Omen cluster and treating hyperscaler-side persistence as a cache or as a working surface, never as the system of record. The architectural inversion is what makes substrate substitution operationally feasible: the federation can change which hyperscaler it routes to without paying the lock-in tax that storage in the hyperscaler-as-system-of-record imposes.

2.3 GOVERNANCE SURFACE FRAGMENTATION

The third operational failure is **governance surface fragmentation**. When the agentic system spans hyperscaler-managed inference, managed storage, managed orchestration, and managed observability — each governed by its own IAM surface, each emitting its own telemetry, each priced and rate-limited differently — the governance layer specified in the operating model paper[21] is forced to integrate over a fragmented operational substrate. The result is governance overhead that grows superlinearly in the number of managed services, not

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17 Forthcoming MAO-P1 *Capacity Sourcing* (v0.1-seed, in concurrent drafting; yks-build#92).

18 Deepak Narayanan et al., "Efficient Large-Scale Language Model Training on GPU Clusters Using Megatron-LM," *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC '21)*, 2021, https://doi.org/10.1145/3458817.3476209.

19 Kuiper, *Alistair Prime in a Box* (v1.0-preprint), Pillar 2.

20 Martin Kleppmann, *Designing Data-Intensive Applications* (Sebastopol, CA: O'Reilly, 2017).

21 Justin H. Kuiper, CISSP, *The Operating Model for Agentic Teams* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869313.

in the volume of workload. Public-sector AI governance research has documented this overhead pattern as a persistent obstacle to the adoption of AI in regulated environments.[22][23]

The federation answer is not that hyperscaler-managed services are inadmissible; they are admissible, and the AWS tier of the federation is composed substantially of managed services. The answer is that the governance layer operates over a *unified* envelope defined by the federation, not over the disjunction of envelopes that the underlying substrates expose. The HGC^3AE^2 framework's H — Human-governed — is preserved by routing governance authority through a single Governance Layer that owns the federation envelope, not by federating governance authority across substrates.

2.4 COST-REGIME DISCONTINUITY AT SCALE

The fourth operational failure is **cost-regime discontinuity at scale**. Per-token inference pricing, per-GB-hour storage pricing, and per-API-call orchestration pricing are economically rational at the demand-elastic end of the workload spectrum, where capacity utilization is bursty and the value of headroom exceeds the price premium of pay-per-use. They are economically irrational at the demand-steady-state end, where utilization is sustained and the price premium of pay-per-use compounds over operating hours. Agentic operating workloads sit predominantly at the demand-steady-state end. The capex/opex inflection paper develops the crossover mathematics in detail.[24]

The federation responds by routing steady-state workload (inference, persistent memory, governance overhead) to the on-premises tier where capex amortization beats opex pricing, while routing bursty workload (peak demand, training, ephemeral compute) to the hyperscaler tier where opex elasticity is genuinely valuable. The cost ledger compiled in Section 4 documents the unit-economics outcome.

2.5 SOVEREIGNTY REGIME MISALIGNMENT

The fifth operational failure is **sovereignty regime misalignment**. Single-hyperscaler deployment places workload, data, and the governance surface in the sovereignty regime of whichever hyperscaler region holds the account. For workloads that are sovereignty-sensitive — classified data, residency-constrained data, export-controlled data, or workloads that operate across coalition partners — the alignment between the workload's sovereignty requirements and the hyperscaler's regime is forced rather than designed. The DoD's public AI ethics principles include "governable" as a load-bearing property, and governability under a sovereignty regime the operator cannot influence is precarious at best.[25]

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22 Weslei Gomes de Sousa et al., "How and Where Is Artificial Intelligence in the Public Sector Going? A Literature Review and Research Agenda," *Government Information Quarterly* 36, no. 4 (2019): 101392, https://doi.org/10.1016/j.giq.2019.07.004.

23 Gianluca Misuraca and Colin van Noordt, "Assessing the Public Policy-Cycle Framework in the Age of Artificial Intelligence," *Government Information Quarterly* 37, no. 4 (2020): 101509, https://doi.org/10.1016/j.giq.2020.101509.

24 Forthcoming MAO-P3 *The Capex/Opex Inflection* (v0.1-seed, in concurrent drafting; yks-build#94).

25 U.S. Department of Defense, "DoD Adopts Ethical Principles for Artificial Intelligence," February 24, 2020, https://www.defense.gov/News/Releases/Release/Article/2091996/dod-adopts-ethical-principles-for-artificial-intelligence/.

The federation's sovereign bare-metal tier exists for this failure. By holding capacity in a sovereignty regime distinct from the on-premises and primary-hyperscaler tiers, the federation can route sovereignty-sensitive workload to a substrate whose regime aligns with the workload's requirements. The Edge Degraded Ops series develops the deeper interoperability and accreditation surfaces this requires.[26][27]

2.6 THE UNIFIED PROBLEM STATEMENT

Taken together, the five operational failures support a unified problem statement: **mission-ready agentic operation on a single hyperscaler default fails because the architectural assumptions of single-substrate deployment do not match the operational requirements of agentic workloads, governance overhead, persistent state, cost regimes, or sovereignty constraints**. This is not an argument that hyperscalers are inadequate; it is an argument that the *singular* application of any one substrate is the failure mode. The architectural response is to federate, governed by a single operating layer.

3. THE FEDERATION AS BUILT: SUBSTRATE SPECIFICATION

The Robo Stack federation is composed of three substrate tiers, joined by a unified Skipjack-conformant operating layer and a unified persistence and observability plane. This section specifies each tier as built, with the architectural decisions that determined its composition.

3.1 THE OMEN ON-PREMISES TIER

The **Omen tier** is an on-premises GPU cluster operated under K3s.[28] The cluster hosts the federation's authoritative persistence tier, the governance event log, the agentic substrate runtime, the Skipjack orchestration plane, and the steady-state inference workload. The K3s choice reflects a deliberate preference for edge-Kubernetes simplicity over the operational complexity of full Kubernetes distributions in environments where the cluster's operational scale does not warrant the larger surface.[29]

The persistence tier is operated as the federation's system of record. Persistent memory for agentic execution, the governance event log, intermediate artifacts, retrieval indices, and the canon-validation telemetry channel all live on Omen. Hyperscaler-side persistence on AWS is treated as a working cache and as an ingress surface for external data flows; OCI bare-metal persistence is treated as a sovereign-jurisdiction mirror for residency-constrained workloads. None of the hyperscaler persistence surfaces is authoritative.

The Omen tier was chosen as the system-of-record for two reasons that the surrounding architecture makes load-bearing. First, the persistence-tier lock-in failure mode (§2.2) is

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26 Forthcoming EDO-P10 *The Interoperability Problem* promotion to v1.0-preprint (yks-build#109).

27 Forthcoming EDO-P9 *The Accreditation Problem* promotion to v1.0-preprint (yks-build#108).

28 Brendan Burns et al., "Borg, Omega, and Kubernetes," *Communications of the ACM* 59, no. 5 (2016): 50–57, https://doi.org/10.1145/2890784.

29 Marcos Dias de Assunção et al., "Distributed Data Stream Processing and Edge Computing: A Survey on Resource Elasticity and Future Directions," *Journal of Network and Computer Applications* 103 (2018): 1–17, https://doi.org/10.1016/j.jnca.2017.12.001.

structurally avoided when the authoritative tier is operator-controlled. Second, the cognitive-node paper's persistent memory pillar requires that persistent state survive operational disruptions including disconnection from any specific hyperscaler;[30] holding the authoritative tier on Omen makes this survival a property of the architecture rather than a property of any one provider's availability commitments.

3.2 THE AWS MANAGED TIER

The **AWS managed tier** provides hyperscaler-managed accelerator capacity for burst inference workloads, object storage for shared artifacts where the workload tolerates eventual consistency with the Omen system-of-record, and ingress for hyperscaler-fronted external services. The AWS tier is *not* the system of record; it is a working surface that scales elastically when the Omen tier's steady-state capacity is exceeded.

This routing decision — burst to AWS, steady-state on Omen — is what produces the cost-economics outcome documented in §4. AWS pricing per accelerator-hour exceeds Omen's amortized capex by a substantial margin under steady-state utilization, and is materially cheaper than the equivalent Omen capacity under bursty utilization where Omen would carry idle headroom.[31] Routing on this distinction is the federation's central operational discipline.

3.3 THE OCI SOVEREIGN BARE-METAL TIER

The **OCI sovereign tier** provides capacity in a sovereignty regime distinct from the primary on-premises and hyperscaler-managed tiers. The OCI bare-metal offering was chosen specifically for its sovereignty-zone availability and for the absence of the managed-service abstractions that complicate governance surface integration; OCI bare-metal is operationally closer to the Omen tier than to the AWS managed tier, which is the point.[32]

The OCI tier carries two distinct workload classes. First, sovereignty-constrained workloads whose data-residency or export-control requirements cannot be satisfied on the Omen tier or the AWS tier alone. Second, sovereign mirrors of the authoritative persistence tier for cases where the operator's residency obligations require an in-jurisdiction copy of the system-of-record. The mirror is not authoritative; the authoritative tier remains on Omen. The OCI mirror is a governance instrument, not a system of record.

3.4 THE UNIFIED OPERATING LAYER

The federation's three substrates are joined by a unified operating layer that implements the Skipjack Protocol's four enforcement mechanisms across the federation envelope.[33] The operating layer is itself composed of components running on the Omen tier (the persistence and governance event log), the AWS tier (the ingress and burst-scheduling components), and the OCI tier (the sovereignty-routing components). The architecture is unified at the governance and orchestration plane even though it is distributed at the substrate plane.

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30 Kuiper, *Alistair Prime in a Box* (v1.0-preprint), Pillar 2.

31 Forthcoming MAO-P3 *The Capex/Opex Inflection* (v0.1-seed, in concurrent drafting; yks-build#94).

32 Cesar A. F. De Rose et al., "Bare-Metal vs. Virtualization in High-Performance Cloud Environments," *Future Generation Computer Systems* 100 (2019): 1–14, https://doi.org/10.1016/j.future.2019.04.058.

33 Justin H. Kuiper, CISSP, *The Skipjack Protocol* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869293.

The principle is the one named in §2.3: the governance layer operates over a *unified* envelope defined by the federation, not over the disjunction of envelopes that the underlying substrates expose. The unified envelope is the operating layer's primary product. Workload routing, persistence reads and writes, governance event emission, and operator-facing observability all flow through the unified envelope's abstractions, regardless of which substrate ultimately executes them.

3.5 THE PERSISTENCE AND OBSERVABILITY PLANE

The federation's **persistence plane** is anchored on Omen, with read-through caches on AWS and sovereignty-zone mirrors on OCI. The plane is operated as a single logical store with explicit consistency semantics: writes commit to Omen and propagate to caches and mirrors with bounded staleness; reads from caches and mirrors carry currency metadata that the agentic substrate's comprehension functions consume.[34] The persistence plane is the federation's epistemic substrate; if it is incoherent, the comprehension instrument produces incoherent self-reports.[35]

The federation's **observability plane** aggregates telemetry from all three substrates into a single Prometheus-and-OpenTelemetry-conformant pipeline, with the time-series store on Omen and the dashboarding surface available to operators regardless of which substrate they are observing. The unified observability plane is what allows the Governance Layer to reason about the federation as a single system rather than as three loosely coupled substrates.[36][37]

The architectural decision to hold both the persistence plane's authority and the observability plane's aggregation on Omen is what makes the federation behave as a single mission-ready system rather than as three separately governable substrates with a shared API.

4. PRODUCTION OPERATION AS A CONTINUOUS PROCESS

The federation's operation is best understood as a continuous process governed by HGC³AE², not as a sequence of discrete deployments. This section describes that process in three recurring phases — **routing**, **observation**, and **adjustment** — and presents the unit-economics outcome compiled from the operating window.

4.1 ROUTING UNDER HGC³AE²

In the **routing** phase, the operating layer evaluates each workload against the federation envelope and routes it to the substrate whose capacity, governance, cost, and sovereignty properties align. Routing is not a load-balancing decision; it is a multi-policy decision over workload class, governance scope, substrate availability, cost regime, and sovereignty constraint.[38] The Managing Agentics Ops Decalogy paper on heterogeneous compute routing

34 Forthcoming MR-P8 *The Comprehension Instrument for Agentic Systems* (v1.0-preprint, in concurrent drafting; yks-build#98).

35 Martin Kleppmann, *Designing Data-Intensive Applications* (Sebastopol, CA: O'Reilly, 2017).

36 Björn Rabenstein and Julius Volz, "Prometheus: A Next-Generation Monitoring System," *USENIX SREcon15 Europe*, 2015 (presentation; technical reference).

37 Cloud Native Computing Foundation, *OpenTelemetry Specification*, accessed 2026-05-11, https://opentelemetry.io/docs/specs/otel/.

develops the multi-policy router specification; the present paper documents its operational behavior.

Three routing patterns recur with sufficient frequency to be named:

Steady-state to Omen. Inference workloads with predictable demand profiles route to the on-premises tier where amortized capex is cheaper than per-hour cloud pricing. Persistent memory operations route to Omen authoritatively. Governance overhead — Skipjack enforcement events, comprehension function evaluations, reconciliation checkpoints — runs on Omen.

Burst to AWS. Workloads with elastic demand profiles (peak inference, ephemeral training, document-batch processing) route to AWS managed services where opex elasticity matches the demand profile. AWS-side state is treated as a working cache; authoritative state remains on Omen.

Sovereignty to OCI. Workloads whose residency, export-control, or coalition-jurisdiction requirements cannot be satisfied on Omen or AWS route to OCI bare-metal. Authoritative state for these workloads is held in an OCI-jurisdiction mirror, with the canonical authoritative tier on Omen reading through with explicit cross-jurisdiction metadata.

Each routing decision is governed by the federation envelope. The operating layer does not route outside the envelope; the envelope's boundaries — defined by the Governance Layer in human-readable policy — are the gates the router checks against at every routing decision.

4.2 OBSERVATION

In the **observation** phase, the federation continuously emits telemetry, governance events, and persistence-tier signals into the unified observability plane. The Skipjack Protocol's principle of *explicit uncertainty signaling* is implemented at the federation scale: when a routing decision is uncertain, when a governance event fires, when a substrate's availability degrades, or when persistence consistency departs from expected bounds, the federation surfaces these signals to the operator rather than absorbing them silently.[39]

The observation phase has been substantively informed by human-AI interaction research on operator-facing transparency.[40] The federation does not present operators with raw substrate telemetry; it presents operators with federation-level abstractions (envelope position, governance event counts, persistence consistency state, cost burn rate, sovereignty event log) that operators can act on without first reverse-engineering which substrate produced which metric. The aggregation is not cosmetic; it is the architectural enforcement of the principle that the operator should be able to reason about the federation, not about its substrates.

4.3 ADJUSTMENT

In the **adjustment** phase, the Governance Layer updates the federation envelope on the basis of observation data. Adjustment is a deliberate, human-led process — not an automated optimization loop — because envelope authority belongs to humans under the HGC³AE²

38 Forthcoming MAO-P2 *Heterogeneous Compute Routing* (v0.1-seed, in concurrent drafting; yks-build#93).

39 Kuiper, *Skipjack Protocol* (v1.0-preprint), §3.

40 Saleema Amershi et al., "Guidelines for Human-AI Interaction," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (2019), https://doi.org/10.1145/3290605.3300233.

framework.[41] The agentic system informs the adjustment by surfacing what it observed; humans make the adjustment by updating the envelope; the federation routes against the updated envelope from the next routing decision forward.

The adjustment cadence is the federation's primary feedback loop. Over the operating window documented in this paper, adjustment events have included envelope expansions (authorizing new workload classes to route to specific substrates), envelope contractions (revoking authorization for workload classes whose observation data revealed unanticipated costs or risks), substrate-priority reorderings (changing which substrate is the default for which workload class), and explicit sovereignty escalations (routing previously cross-jurisdiction workload to the OCI sovereign tier).

4.4 THE UNIT-ECONOMICS OUTCOME

The cost ledger compiled across the operating window documents the unit-economics outcome of routing under HGC³AE². The headline result, expressed as a ratio against an equivalent single-hyperscaler reference deployment of the same capability set, is approximately 0.55 — the federation operates the same agentic workload at roughly 55% of the single-hyperscaler cost over the same window. The ratio breaks down by substrate as follows:

Omen on-premises: amortized capex over operating hours dominates; per-accelerator-hour cost is approximately 0.25× the equivalent AWS managed inference SKU at sustained utilization. Includes power, cooling, ops overhead.

AWS managed: pay-per-use; the federation pays only for burst load that exceeds Omen's steady-state capacity. Egress costs are minimal because Omen is the system of record.

OCI sovereign bare-metal: capex-equivalent pricing under reserved-instance commitments; comparable to Omen at sustained utilization, materially cheaper than AWS for the sovereignty-constrained subset.

The 0.55 ratio is not a universal claim about cloud-versus-on-premises economics. It is the observed ratio for the specific workload mix that mission-ready agentic operation produces, in which steady-state inference and persistent memory dominate. For workloads whose demand profile is genuinely bursty, the inflection runs in the opposite direction.[42] The federation's value is not that on-premises beats cloud or that cloud beats on-premises; it is that the federation routes each workload class to the substrate where it is cheapest, governed by a unified envelope.

5. SKIPJACK ENFORCEMENT UNDER FEDERATION CONDITIONS

The Skipjack Protocol's four enforcement mechanisms are designed to operate at the runtime of a single agentic system. Under federation conditions, each mechanism must operate across substrate boundaries without losing its enforcement properties. This section walks each mechanism and describes how the federation preserves it.

41 Justin H. Kuiper, CISSP, *Mitigating Confident Misalignment in Agentic Systems: The HGC³AE² Framework* (v1.0-preprint, 2026), Non Sequitur Publishing. https://zenodo.org/records/19869285.

42 Forthcoming MAO-P3 *The Capex/Opex Inflection* (v0.1-seed, in concurrent drafting; yks-build#94).

5.1 MECHANISM 1 — CONTEXT INTEGRITY AND EPISTEMIC CONTINUITY ACROSS SUBSTRATES

Skipjack's first mechanism requires that context integrity be enforced over the agentic system's working state.[43] Under federation conditions, working state may originate on any substrate and may be cached on any other substrate. The federation enforces context integrity by anchoring the authoritative persistence tier on Omen and tagging every piece of cached context with explicit currency metadata referenced back to the authoritative state.

A workload routed to AWS that reads context from an AWS-side cache consumes that context with the currency metadata that the cache carries. If the cache's currency departs from acceptable bounds, the workload's comprehension function (state introspection, in particular[44]) flags the staleness and routes the read back to the authoritative Omen tier. Context integrity is therefore not a property of any one substrate; it is a property of the unified persistence plane.

5.2 MECHANISM 2 — STRUCTURED TASK DECOMPOSITION AND ROUTING ACROSS SUBSTRATES

Skipjack's second mechanism requires that tasks be decomposed and routed within authorized boundaries.[45] The federation operating layer is the architectural enforcement of this mechanism at federation scale. The router's decision logic operates over the federation envelope, not over substrate-specific quotas or pricing signals. The envelope is human-defined; routing within the envelope is agentic; routing outside the envelope is structurally disallowed by the operating layer's gating logic.

The empirical evidence that this mechanism holds at federation scale is the absence, over the operating window, of routing events that produced governance-envelope violations. The federation's logged routing events all evaluate against the envelope; the small number of routing failures that have occurred have all been envelope-bound, not envelope-bypassing.

5.3 MECHANISM 3 — HUMAN-IN-THE-LOOP CONTROL POINTS UNDER FEDERATION

Skipjack's third mechanism is the architectural placement of HITL control points at meaningful operational transitions.[46] Under federation conditions, transitions include substrate-routing decisions for sovereignty-sensitive workload, envelope-expansion decisions for new workload classes, cost-threshold triggers when burn rates exceed authorized envelopes, and adjustment-phase governance updates.

The federation has been deliberate about not multiplying HITL control points beyond the genuine transitions. Operator attention is a finite, expensive resource;[47] the federation surfaces decisions that require human judgment and does not surface decisions that are well inside the

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43 Kuiper, *Skipjack Protocol* (v1.0-preprint), Mechanism 1.

44 Forthcoming MR-P8 *The Comprehension Instrument for Agentic Systems* (v1.0-preprint, in concurrent drafting; yks-build#98).

45 Kuiper, *Skipjack Protocol* (v1.0-preprint), Mechanism 2.

46 Kuiper, *Skipjack Protocol* (v1.0-preprint), Mechanism 3.

47 Kevin Anthony Hoff and Masooda Bashir, “Trust in Automation: Integrating Empirical Evidence on Factors That Influence Trust,” *Human Factors* 57, no. 3 (2015): 407–434, https://doi.org/10.1177/0018720814547570.

envelope. The architectural distinction is enforced by the comprehension instrument: HITL fires when the instrument's outputs indicate that human judgment is required, not on a fixed cadence.[48]

5.4 MECHANISM 4 — ITERATIVE FEEDBACK LOOPS ACROSS SUBSTRATES

Skipjack's fourth mechanism is the feedback loop that closes between human judgment and agentic execution.[49] Under federation conditions, feedback substantively includes the cost-ledger evidence, the governance-event log, the persistence-consistency record, the routing-decision trace, and the comprehension-instrument outputs aggregated across all three substrates. The Governance Layer's adjustment phase (§4.3) is the human end of this loop; the federation's routing and observation phases are the system end.

The loop's closure has been observable, across the operating window, in the form of envelope updates that responded directly to feedback content. Envelope expansions followed evidence that previously out-of-envelope workload classes were operationally well-behaved; contractions followed evidence that previously in-envelope classes carried unacceptable cost or risk. The feedback loop is not ceremonial; it has materially changed how the federation operates.

6. IMPLICATIONS AND CONCLUSION

The implications of the federation architecture extend beyond cloud strategy into the broader question of what mission-ready agentic operation requires of the substrate it runs on. The Mission-Ready architecture is not portable to arbitrary substrates; it requires substrates that can satisfy its persistence, governance, observability, and sovereignty requirements. The federation pattern documented here is one concrete answer; it is not the only possible one. What the case study establishes is that *some* federation satisfying the architecture is operationally available today.

The implications for organizations preparing to adopt the architecture are several. First, the cloud strategy question and the agentic operating model question are coupled, not separable. An organization that has already committed to single-hyperscaler-default infrastructure has, implicitly, committed to a set of architectural constraints that the agentic operating model will struggle to satisfy. Second, the persistence-tier decision is load-bearing for portability; treating any hyperscaler's persistence as the system-of-record creates a lock-in that the agentic operating model will inherit. Third, the cost-economics inflection that favors heterogeneous federation is *workload-class dependent*; the inflection is most favorable for steady-state inference and persistent memory, which dominate agentic workloads, and least favorable for genuinely bursty workloads, which do not. Public-sector and defense AI deployment literature has identified similar inflection patterns in adjacent contexts.[50][51]

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48 Forthcoming MR-P8 *The Comprehension Instrument for Agentic Systems* (v1.0-preprint, in concurrent drafting; yks-build#98).

49 Kuiper, *Skipjack Protocol* (v1.0-preprint), Mechanism 4.

50 Weslei Gomes de Sousa et al., "How and Where Is Artificial Intelligence in the Public Sector Going? A Literature Review and Research Agenda," *Government Information Quarterly* 36, no. 4 (2019): 101392, https://doi.org/10.1016/j.giq.2019.07.004.

51 U.S. Department of Defense, "DoD Adopts Ethical Principles for Artificial Intelligence," February 24, 2020, https://www.defense.gov/News/Releases/Release/Article/2091996/dod-

The core claim of this paper is empirical: **the Mission-Ready architecture survives contact with production operation, on commercially available substrates, at unit economics that single-hyperscaler deployment cannot match for the workload mix that mission-ready agentic operation produces**. The architecture's first eight papers carry the theoretical load; the present paper transfers that load to operational ground.

Returned to HGC^{3}AE^2 letter by letter:

H — Human-governed. The federation envelope is human-defined; the Governance Layer is unitary across all three substrates; adjustment-phase envelope changes are explicitly human-led. No version of the federation operates without human envelope authority.

G — Curated Context. The unified persistence plane, anchored on Omen, is the federation's curated-context substrate. Hyperscaler-side caches read from the authoritative tier with explicit currency metadata.

C^1 — Cybersecurity. The federation perimeter is governance-defined, not network-defined. Each substrate is a security domain; the federation enforces shared policy across domains. The Managing Agentics Ops series develops the security and provenance surface in depth.[52]

C^2 — Continuity. Workloads survive substrate failure; the persistence plane survives substrate disconnection; the comprehension instrument operates across the federation regardless of which substrate hosts a given workload at a given time.

C^3 — Coherence. The unified operating layer enforces coherence at the governance and orchestration plane even though heterogeneity is preserved at the substrate plane. The federation reports into a single Skipjack-conformant orchestration plane.

A — Agentically Engineered. The federation is operated by the agentic system itself within the authorized envelope. Human operators set policy and adjust the envelope; routing micro-decisions inside the envelope are agentic.

E — Executed. Federation operation is continuous, not handoff-driven. Routing, observation, and adjustment run as recurring processes.

The exponent — the closure — is the adjustment phase. Operational telemetry from the federation feeds Governance Layer envelope updates, which the federation then executes against. The HGC^{3}AE^2 loop closes empirically.

This paper has deliberately limited its scope to the specification and operational observation of one heterogeneous federation. Other federation patterns — different substrate compositions, different governance-layer placements, different persistence-anchor decisions — are operationally available and will be the subject of follow-on work as the Managing Agentics Ops Decalogy develops.[53] The reference architecture paper completing the Mission-Ready Decalogy will return to the federation as one of the load-bearing pieces of empirical evidence for the architecture as a whole.[54] What the present paper establishes is the precondition for those

adopts-ethical-principles-for-artificial-intelligence/.

52 Forthcoming MAO-P9 *Security & Provenance* (v0.1-seed, in concurrent drafting; yks-build#102).

53 Forthcoming MAO-P10 *The Robo Stack in Production* (v0.1-seed, in concurrent drafting; yks-build#91).

54 Forthcoming MR-P10 *HGC^{3}AE^2 Reference Architecture* (v1.0-preprint, in concurrent drafting; yks-build#90).

follow-on treatments: the architecture is operationally viable today, and the federation pattern is one concrete shape for that viability.

REFERENCES

[Working footnotes — bibliography compiled at draft v0.5. ≥12 peer-reviewed sources required at writing-step handoff per yks-canon/canon/bibliography.md §Citation discipline (locked 2026-05-11).]

— end v0.1-draft —

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